

# KON 313

# Feedback Control Systems

Design via Root Locus

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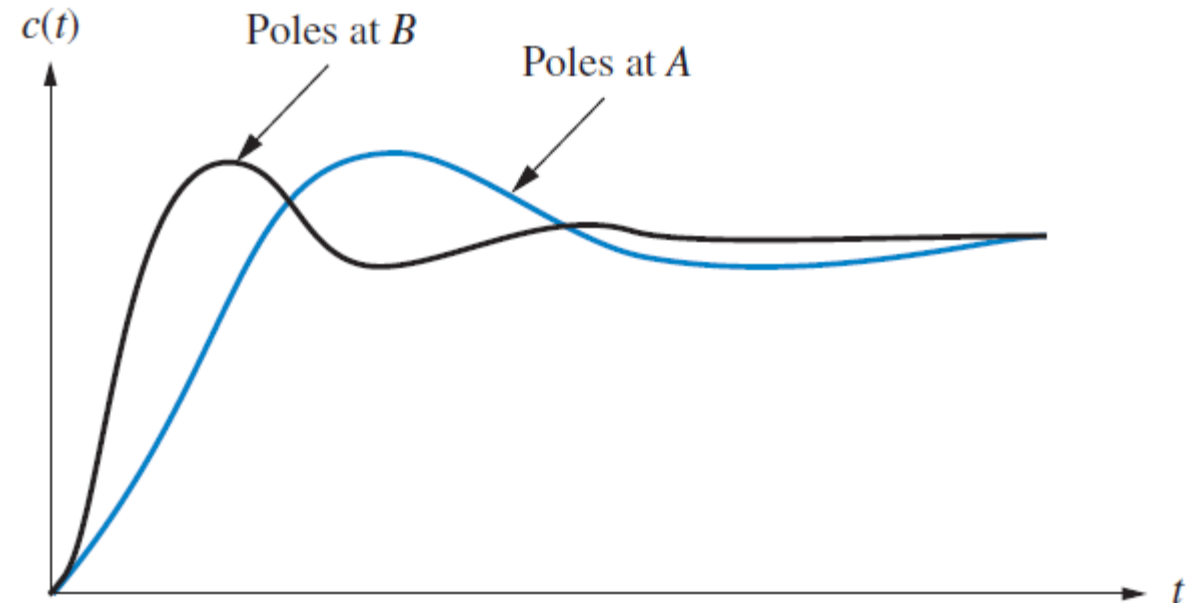
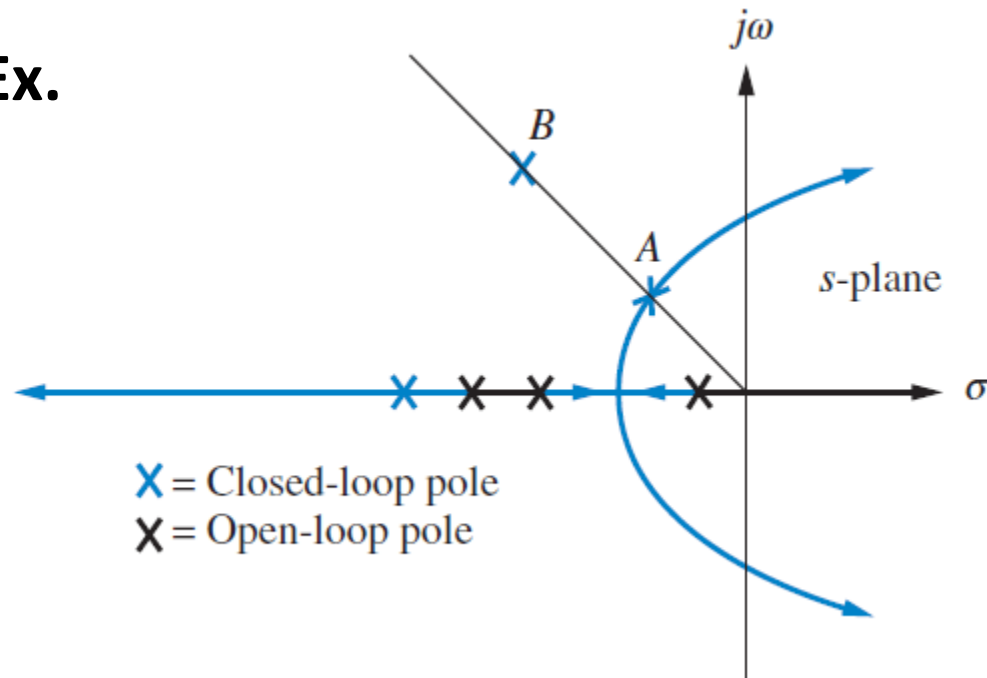
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# Introduction

**Recall. Improving the Transient Response.** Setting the **loop gain** at a particular value, we can obtain the transient response specification dictated by the poles at that point on the root locus. This way, we are **limited to those responses that exist along the root locus**.

**Ex.**



The goal is to speed up the response at A to that of B, without affecting the percent overshoot. This cannot be accomplished by a simple gain adjustment, since point B does not lie on the root locus.

# Introduction

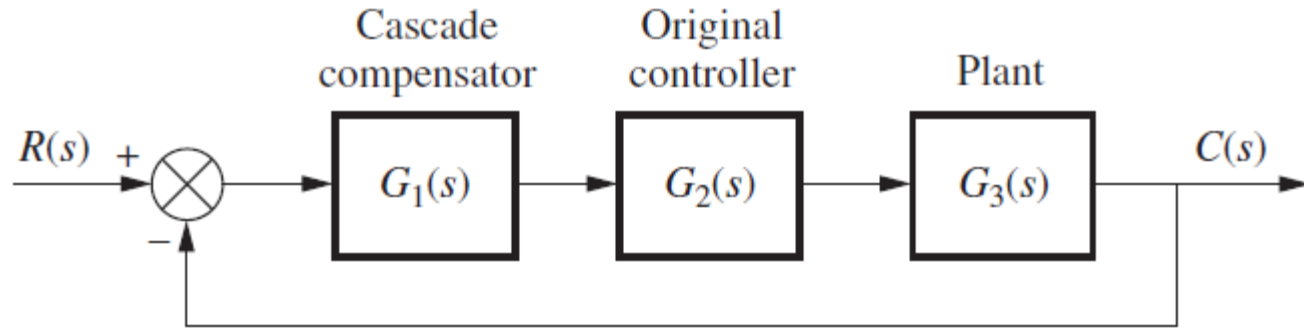
## Recall. Improving Steady-State Error.

- Steady-state error can be improved by adding an open-loop pole at the origin (an **integrator**) in the forward path, thus increasing the system type and driving the associated steady-state error to zero.
- The higher the gain, the smaller the steady-state error (but the larger the percent overshoot).

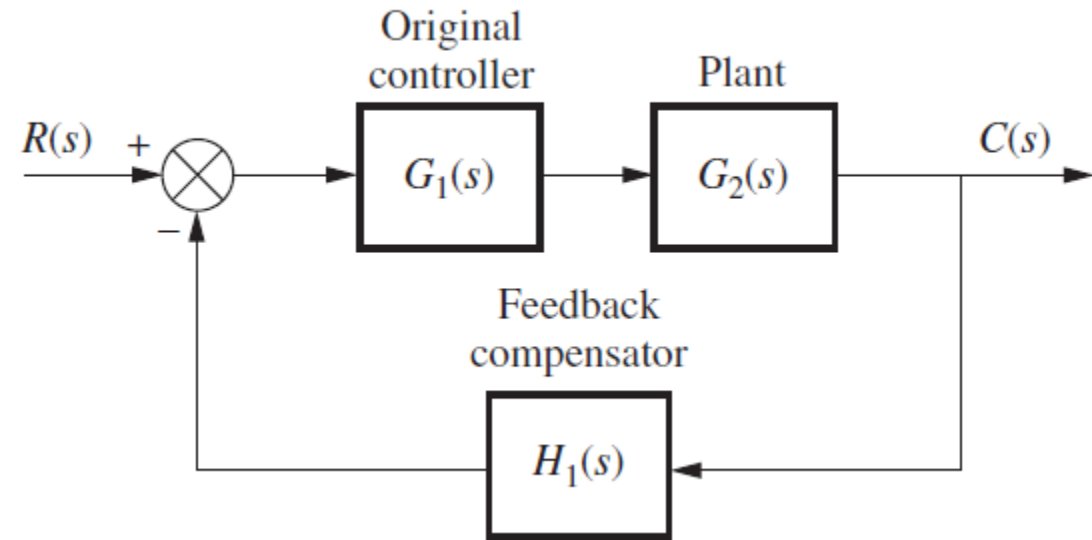
# Introduction - Configurations

Two configurations of compensation are covered here:

**cascade** compensation



**feedback** compensation



Both methods change the **open-loop poles and zeros**, thereby creating a **new root locus** that goes through the **desired closed-loop pole location**.

# Introduction – Compensator Categories

## Active vs Passive compensator.

- Active: implemented with active networks, which, in the case of electric networks, require the use of active **amplifiers** and possible **additional power sources**.
- Passive: is implemented with passive elements such as resistors and capacitors.

The choice between them revolves around cost, weight, desired performance, transfer function, and the interface between the compensator and other hardware.

**Ideal compensators.** Compensators that use **pure integration** or **pure differentiation**. They must be implemented with **active** networks.

**Overall.** All Cascade compensators covered here are

- P (**P**roportional), I (**I**ntegral), D (**D**erivative) control systems
- PI (P-plus-I), PD (P-plus-D), PID (P-plus-I-plus-D)
- Lead. Lag. Lag-Lead

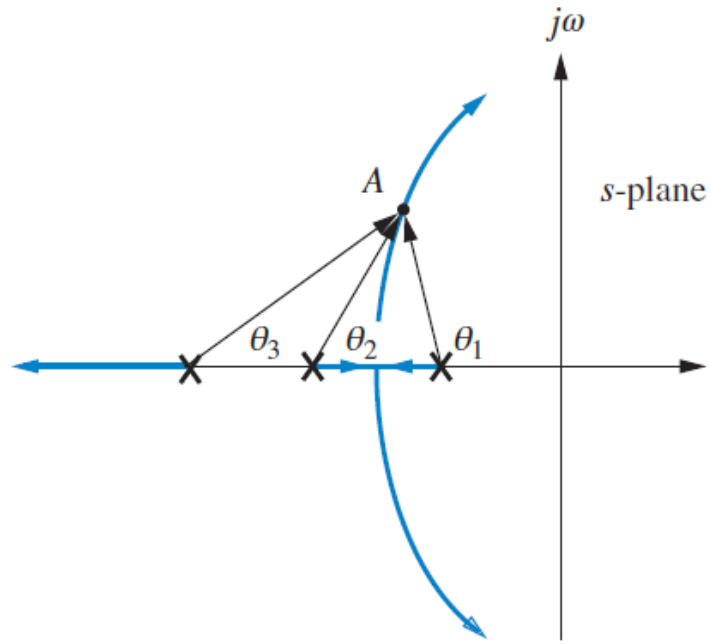
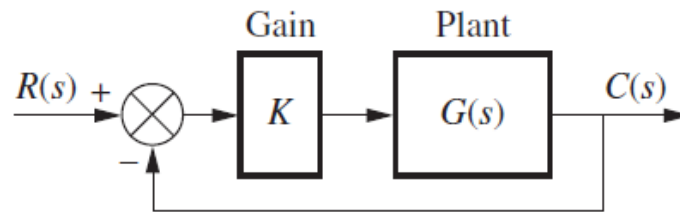
# Improving Steady-State Error via Cascade Compensation

- The objective of this design is to improve the steady-state error **without appreciably affecting the transient response**.
- There are two techniques:
  1. **Proportional-plus-integral (PI) controller. Ideal integral compensation.** uses a **pure integrator** in forward-path; thus increases the system type and reduces the error to **zero**; the compensator must be implemented with **active networks**, such as amplifiers.
  2. **Lag compensator.** places the pole **near the origin**; does not drive the steady-state error to zero, does yield a **measurable reduction in steady-state error**; does have the advantage that it can be implemented with a less expensive **passive network**; the name comes from its frequency response.

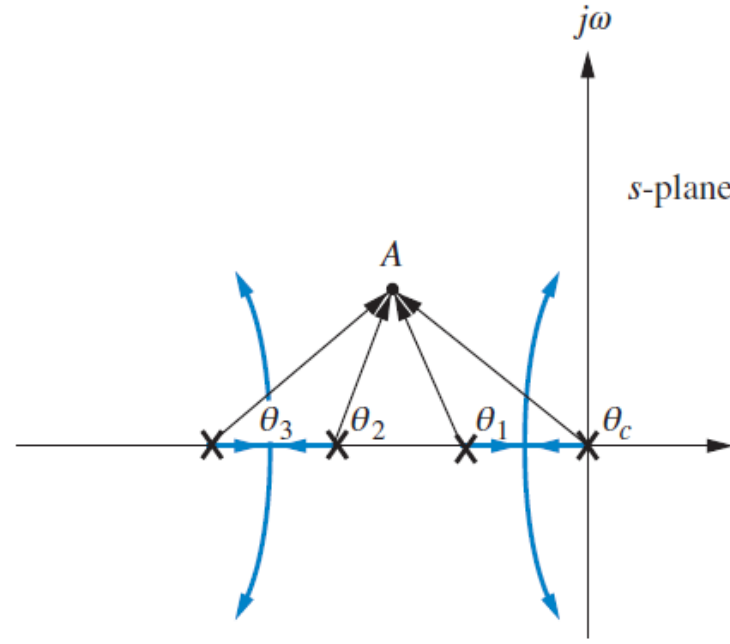
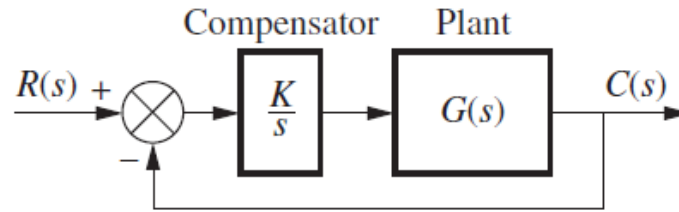
# Improving Steady-State Error via Cascade Compensation

## Ideal Integral Compensation (PI)

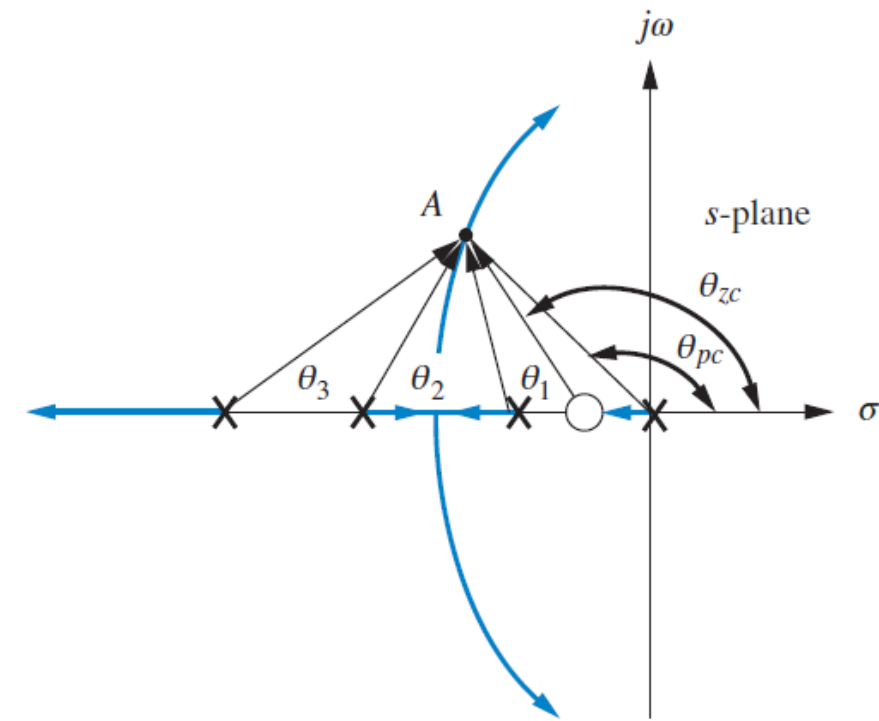
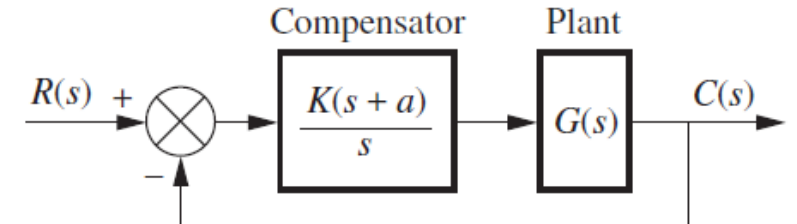
A compensator with a pole at the origin and a zero close to the pole



$$-\theta_1 - \theta_2 - \theta_3 = (2k+1)180^\circ$$



$$-\theta_1 - \theta_2 - \theta_3 - \theta_c \neq (2k+1)180^\circ$$



$$-\theta_1 - \theta_2 - \theta_3 - \theta_{pc} + \theta_{zc} \cong (2k+1)180^\circ$$



# Improving Steady-State Error via Cascade Compensation

## Ideal Integral Compensation (PI)

Placing an **open-loop pole at the origin** increases the system type by one and so can improve the steady-state error.

The aim is to improve the  $e_{ss}$  **without affecting** the transient response.

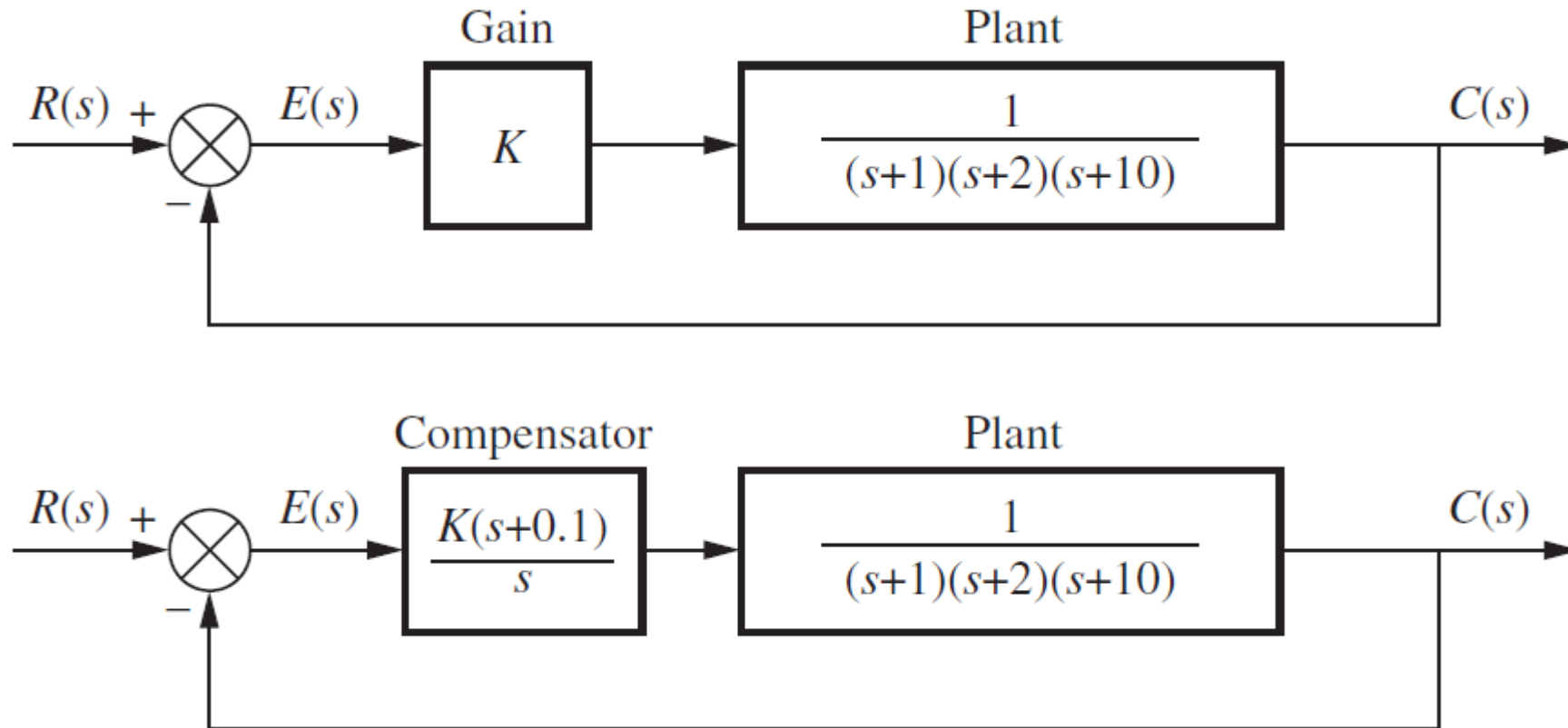
Thus, we also add a **zero close to the origin**, so,

- The **angular** contribution of the compensator zero and pole **cancel out**.
- The required **gain** at the dominant pole is about the same as before compensation, since the ratio of lengths from the compensator pole and zero is approximately unity.

# Improving Steady-State Error via Cascade Compensation

## Ideal Integral Compensation (PI)

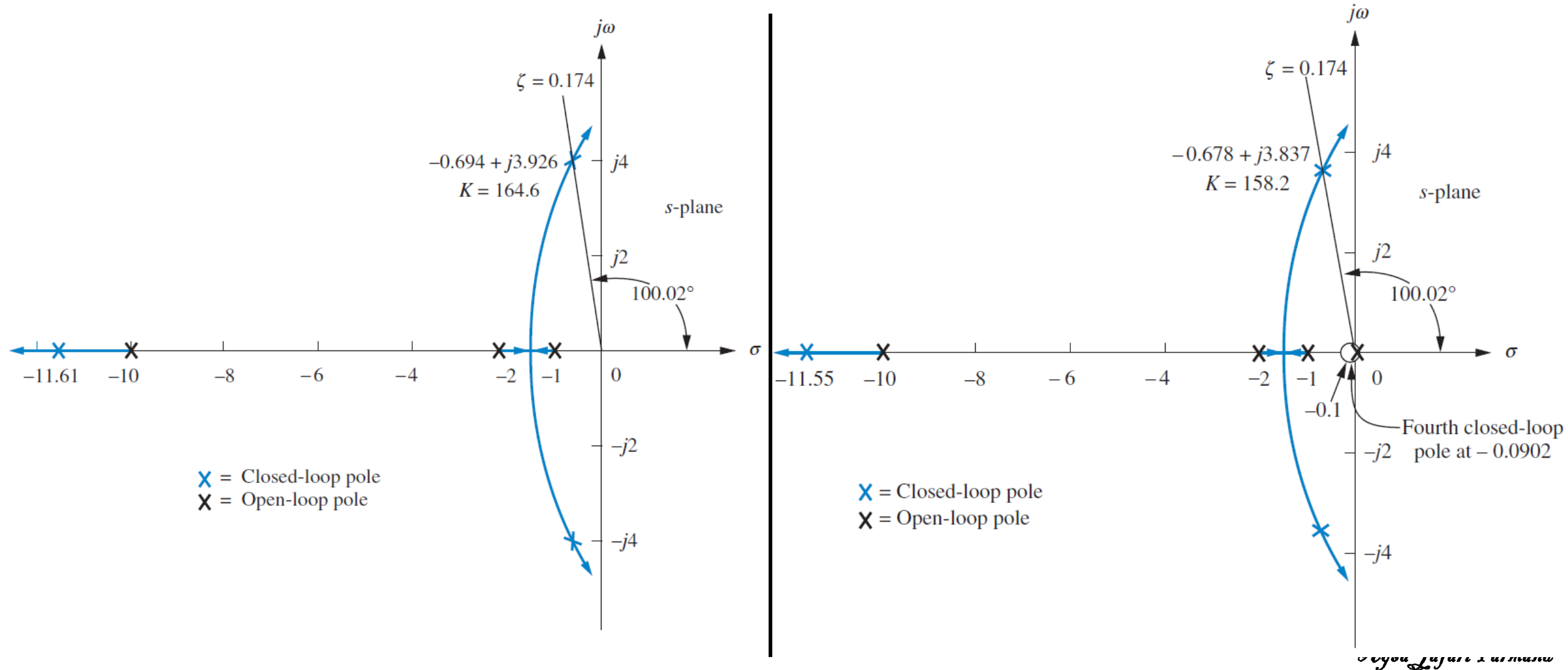
**Ex.** Given the following system, operating with a damping ratio of 0.174, show that the addition of the PI compensator below reduces the  $e_{ss}$  to zero for a step input without appreciably affecting transient response.



# Improving Steady-State Error via Cascade Compensation

## Ideal Integral Compensation (PI)

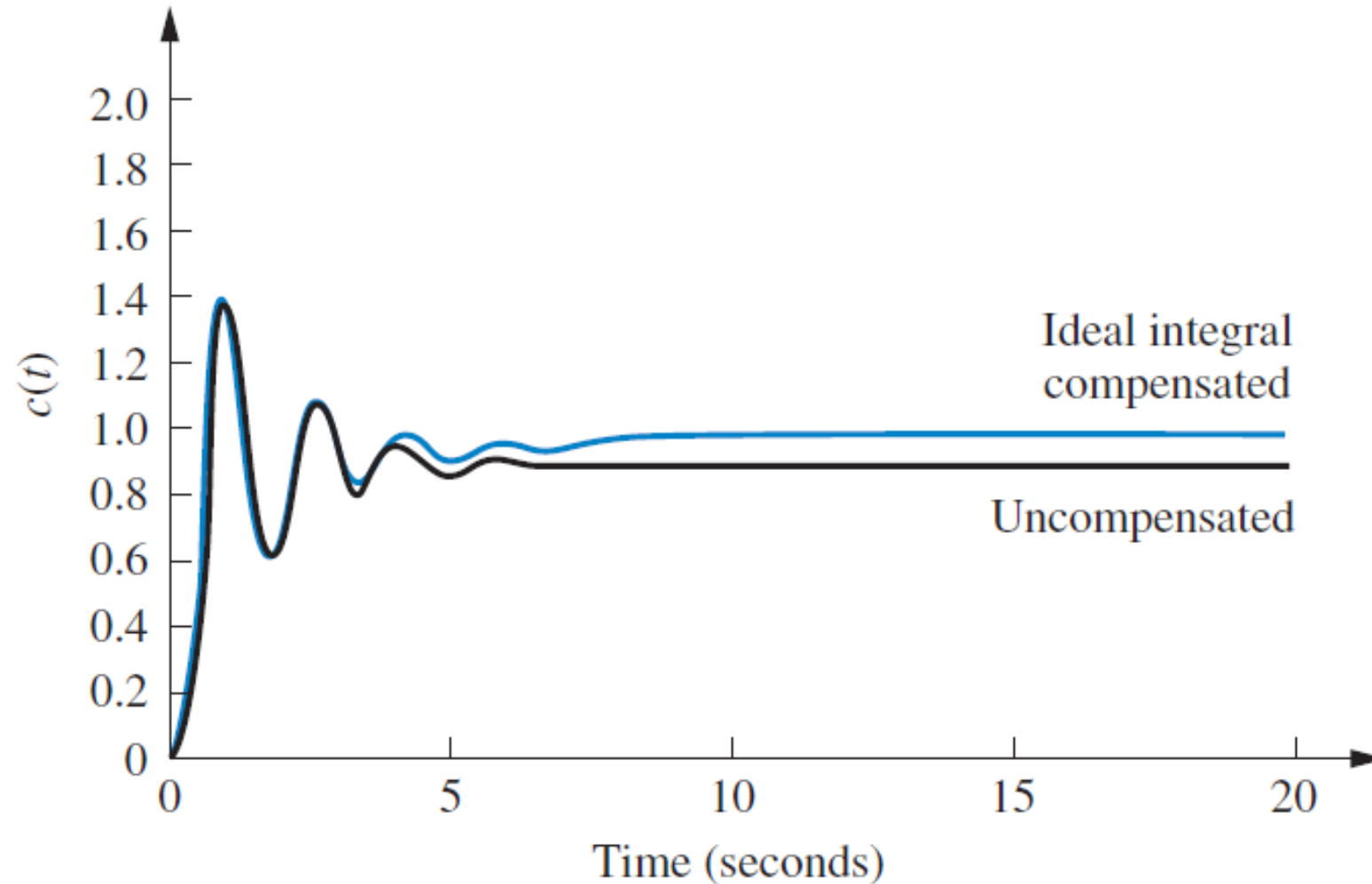
Ex. Cont.



# Improving Steady-State Error via Cascade Compensation

## Ideal Integral Compensation (PI)

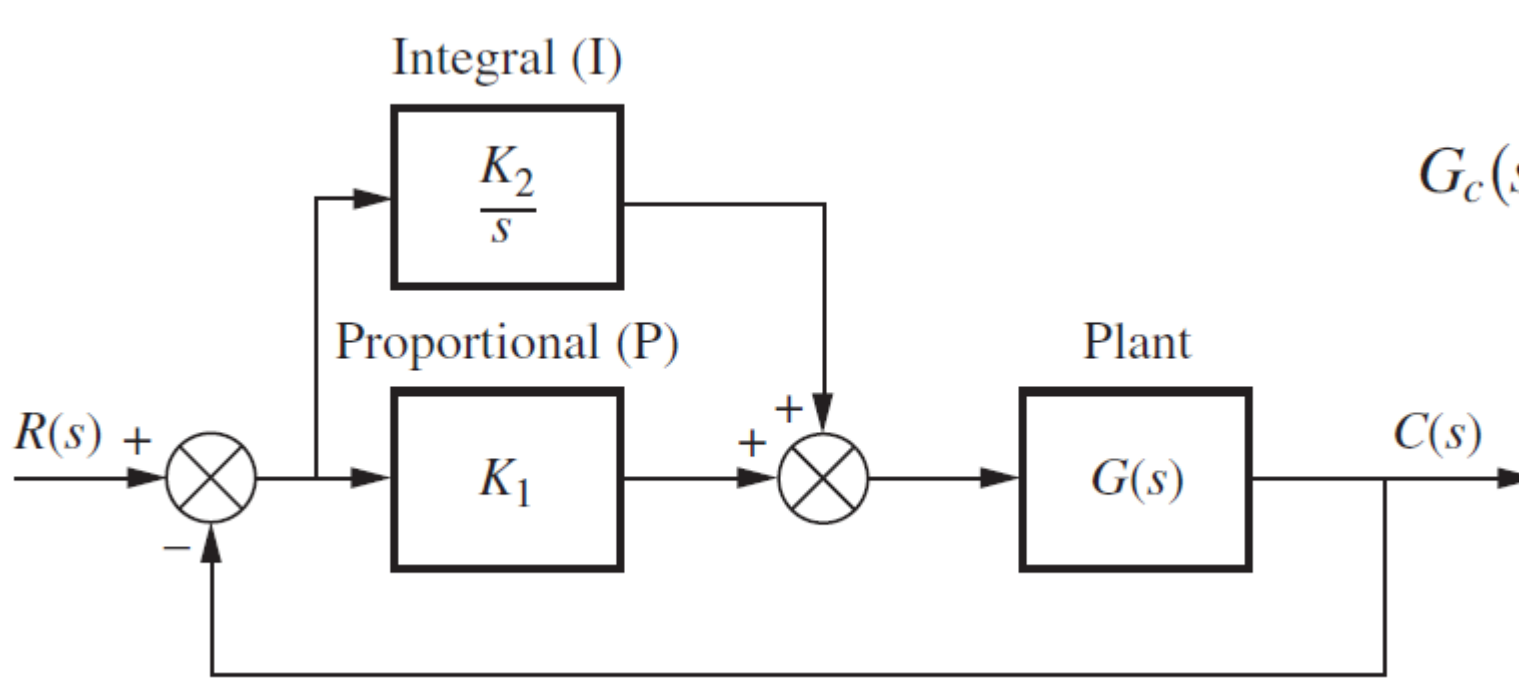
Ex. Cont.



# Improving Steady-State Error via Cascade Compensation

## Ideal Integral Compensation (PI) - Implementation Method

- The value of the zero can be adjusted by varying  $K_2/K_1$ .
- The error and the integral of the error are fed forward to the plant,  $G(s)$ .



$$G_c(s) = K_1 + \frac{K_2}{s} = \frac{K_1 \left( s + \frac{K_2}{K_1} \right)}{s}$$

# Improving Transient Response via Cascade Compensation

- The objective is to design a response that has a **desirable percent overshoot** and a **shorter settling time** than the uncompensated system.
- There are two techniques:
  1. **Proportional-plus-derivative (PD) controller. Ideal derivative compensation.** a **pure differentiator** is added to the forward path; the result is the addition of a **zero** to the forward-path transfer function; requires an **active** network for its realization; differentiation is a noisy process (differentiating high-frequency noise yields a large, unwanted signal);
  2. **Lead Compensator.** It approximates differentiation with a **passive** network by adding a **zero and a more distant pole** to the forward-path transfer function; the name comes from its frequency response.

# Improving Transient Response via Cascade Compensation

## Ideal Derivative Compensation (PD)

The transient response of a system can be selected by choosing an appropriate closed-loop pole location on the  $s$ -plane.

- If this point is on the root locus, then a simple **gain adjustment** is all that is required in order to meet the transient response specification.
- If the closed-loop pole location is not on the root locus, then the root locus must be **reshaped** so that the compensated (new) root locus goes through the selected closed-loop pole location. To accomplish this task, poles and zeros can be added in the forward path to produce a new open-loop function whose root locus goes through the design point on the  $s$ -plane.

One way to speed up the original system is to add a single zero to the forward path. This can be represented by a compensator whose transfer function is

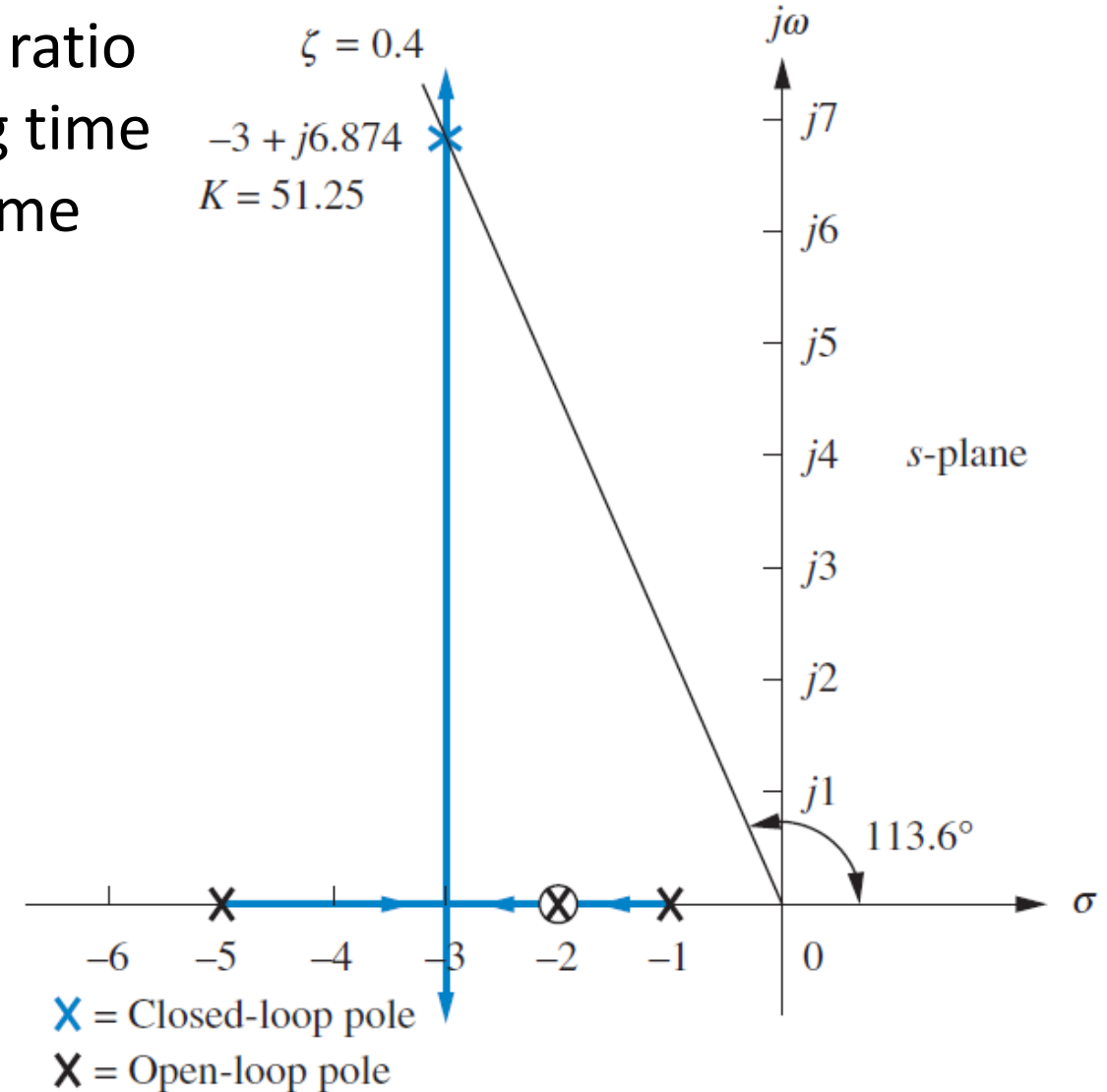
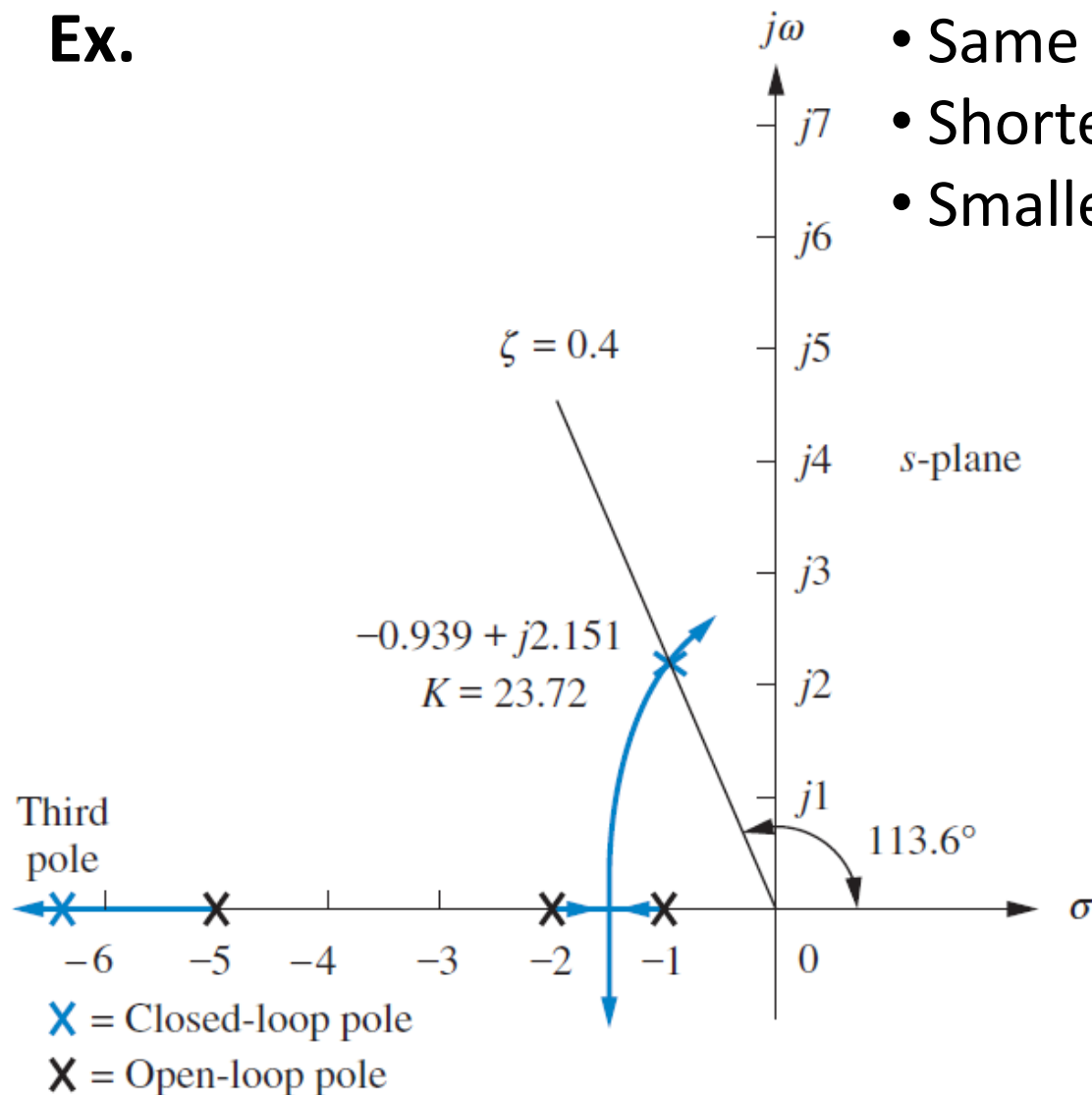
$$G_c(s) = s + z_c$$

# Improving Transient Response via Cascade Compensation

## Ideal Derivative Compensation (PD)

Ex.

- Same damping ratio
- Shorter settling time
- Smaller peak time



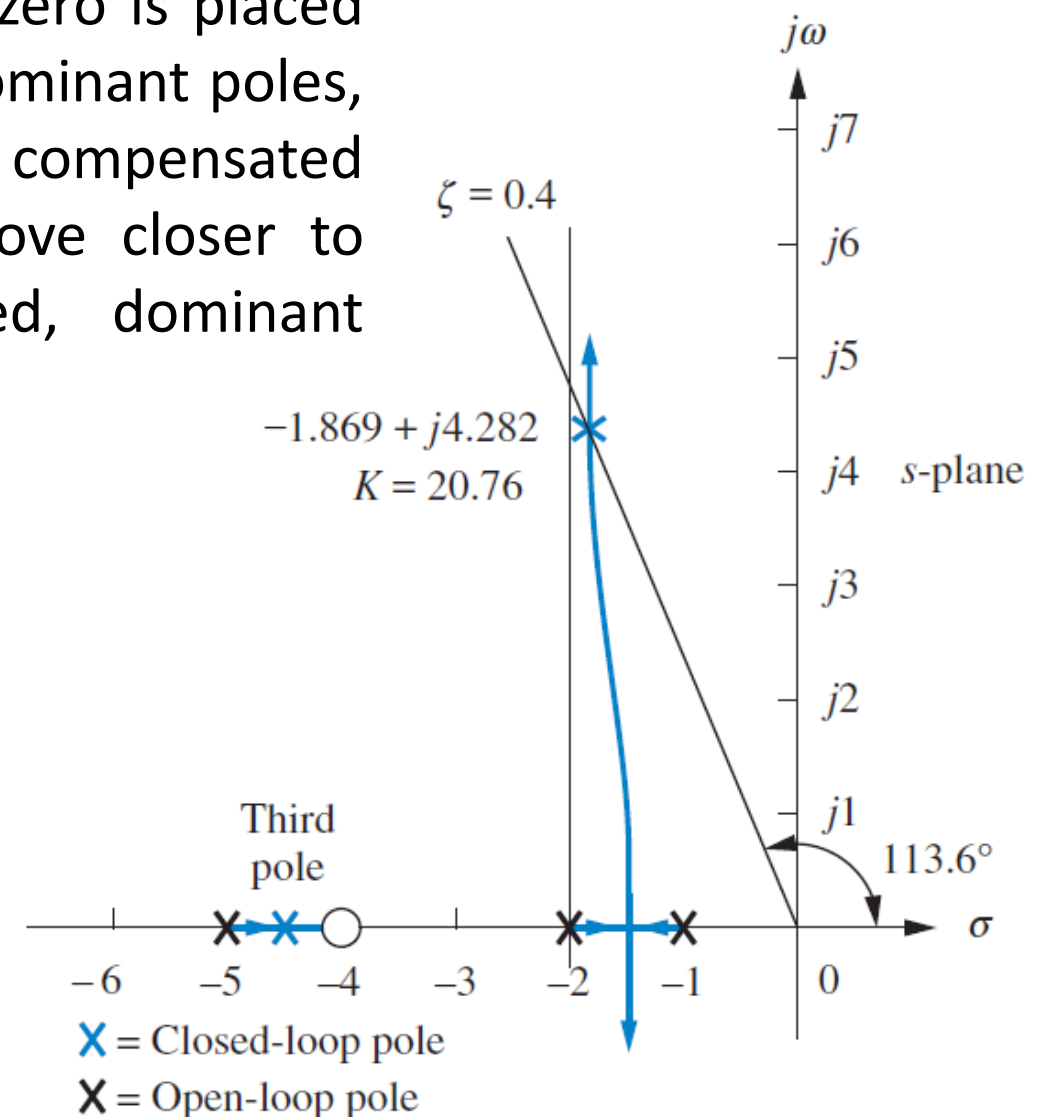
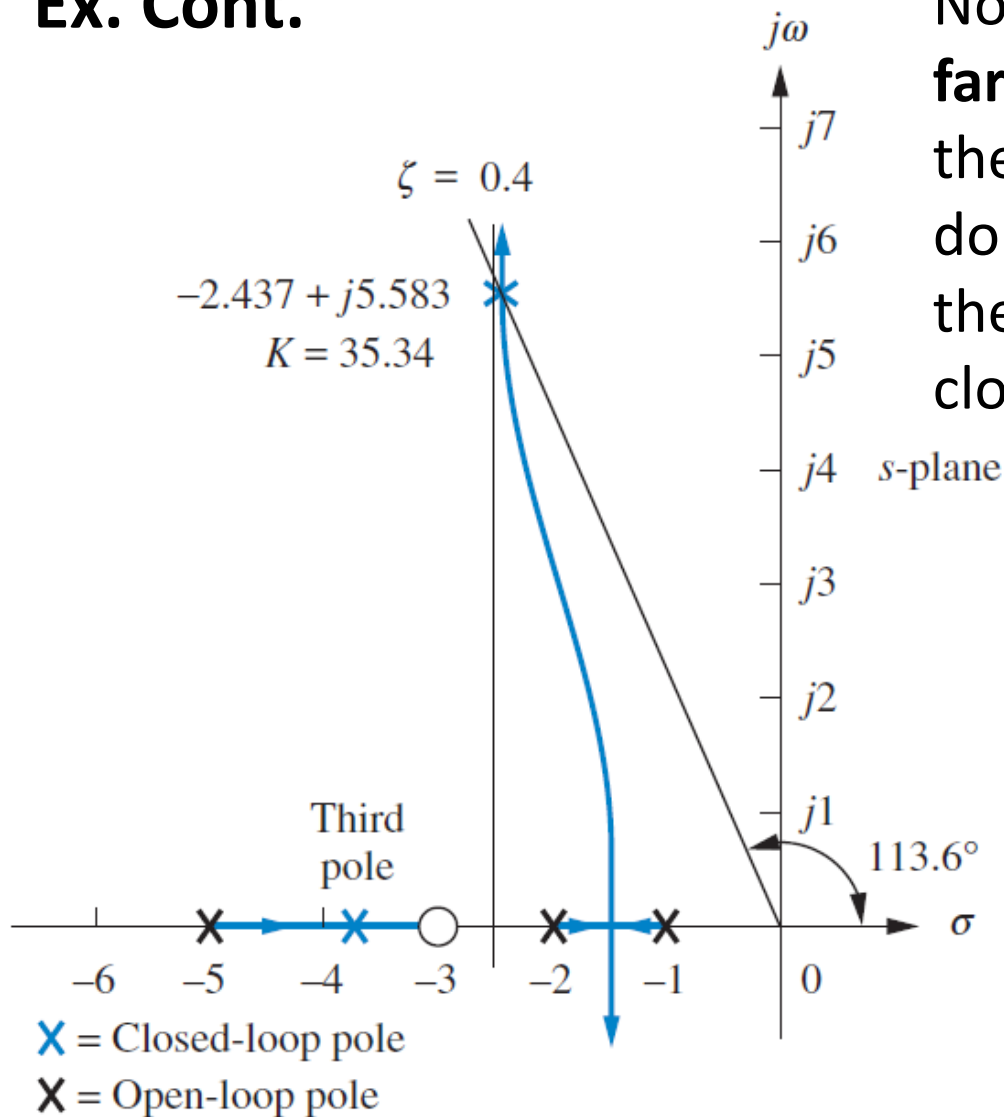


# Improving Transient Response via Cascade Compensation

## Ideal Derivative Compensation (PD)

Ex. Cont.

Notice that as the zero is placed **farther** from the dominant poles, the closed-loop, compensated dominant poles move closer to the uncompensated, dominant closed-loop poles.

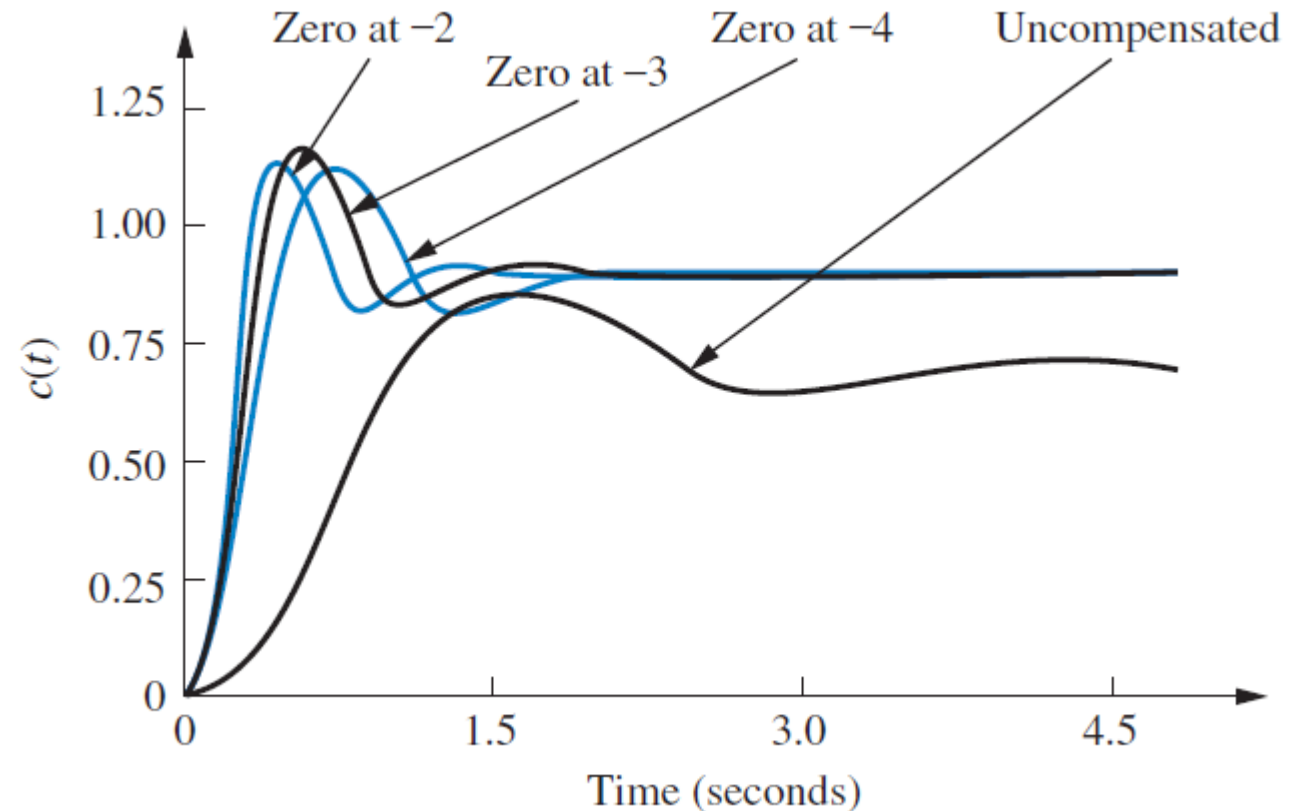


# Improving Transient Response via Cascade Compensation

## Ideal Derivative Compensation (PD)

**Ex. Cont.** In summary, farther zero to the origin yield slower response; In all cases compensation shortened the response time (settling time and peak time) while keeping the percent overshoot the same.

An added benefit is the improvement in the  $e_{ss}$ , even though lag compensation was not used. The reader **must not** assume that, in general, improvement in transient response always yields an improvement in steady-state error.



# Improving Transient Response via Cascade Compensation

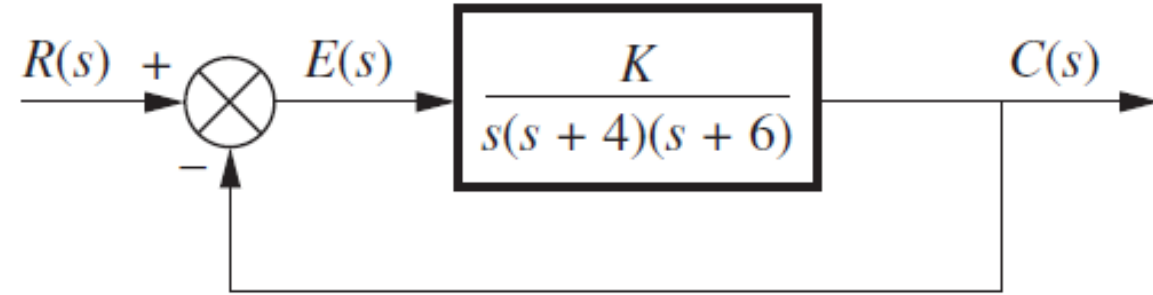
## Ideal Derivative Compensation (PD) – Design Approach

- Basically, we will evaluate the **sum of angles** from the open-loop poles and zeros to a design point that is the closed-loop pole that yields the desired transient response.
- The difference between  $180^\circ$  and the calculated angle must be the angular contribution of the compensator zero.
- Trigonometry is then used to locate the position of the zero to yield the required difference in angle.

# Improving Transient Response via Cascade Compensation

## Ideal Derivative Compensation (PD)

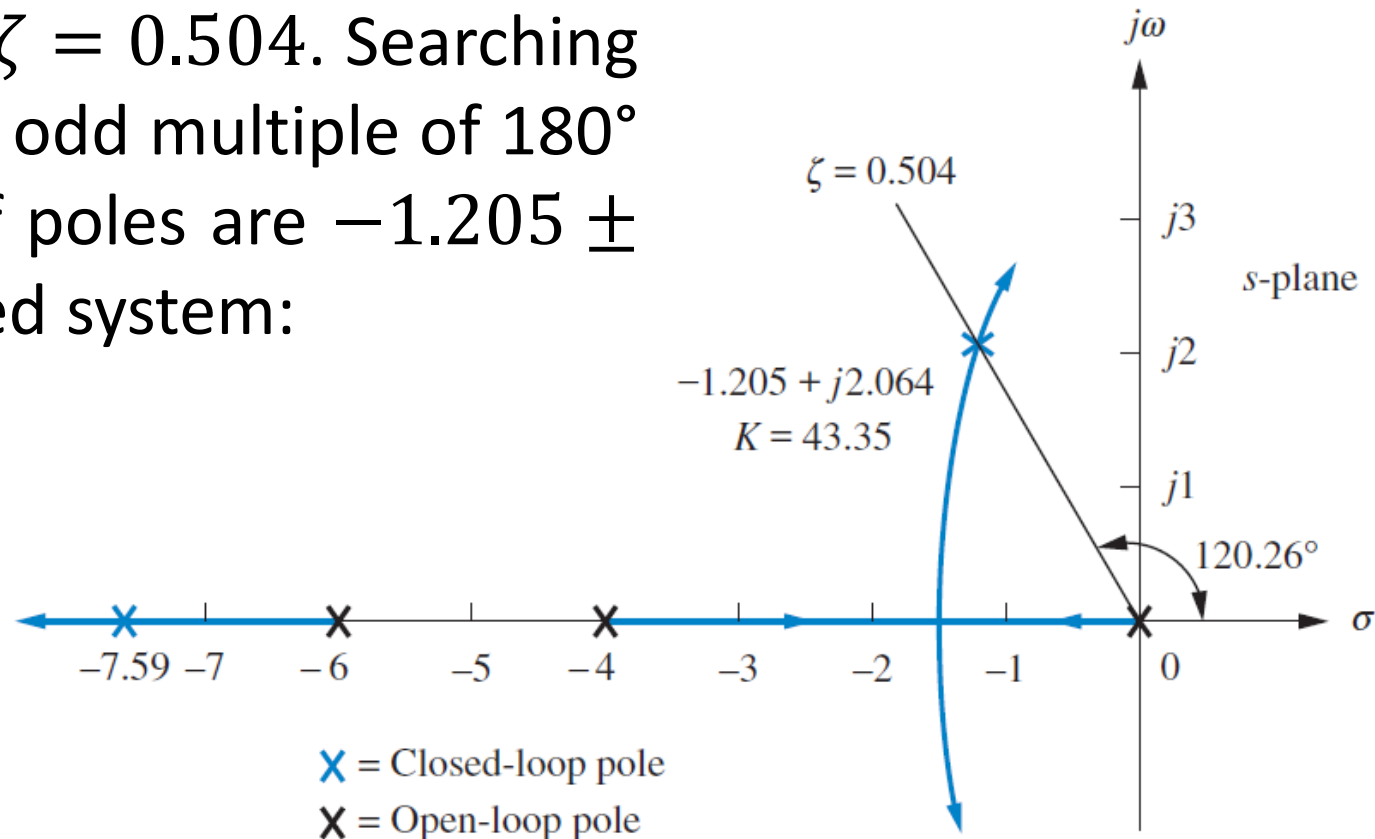
**Ex.** Design a PD compensator to yield a 16% overshoot, with a threefold reduction in settling time.



**Soln.**  $OS\% = 16\%$  is equivalent to  $\zeta = 0.504$ . Searching along that damping ratio line for an odd multiple of  $180^\circ$  the dominant, second-order pair of poles are  $-1.205 \pm j2.064$ . Thus, for the uncompensated system:

$$T_s = \frac{4}{\zeta \omega_n} = \frac{4}{1.205} = 3.320$$

We must check if the second-order approximation is **valid** by finding the third pole and justifying.



# Improving Transient Response via Cascade Compensation

## Ideal Derivative Compensation (PD)

**Soln. Cont.** First we find the location of the compensated system's dominant poles. The new settling time will be 1.107. Therefore, the real part of the compensated system's dominant, second-order pole is

$$\sigma = \frac{4}{T_s} = \frac{4}{1.107} = 3.613$$

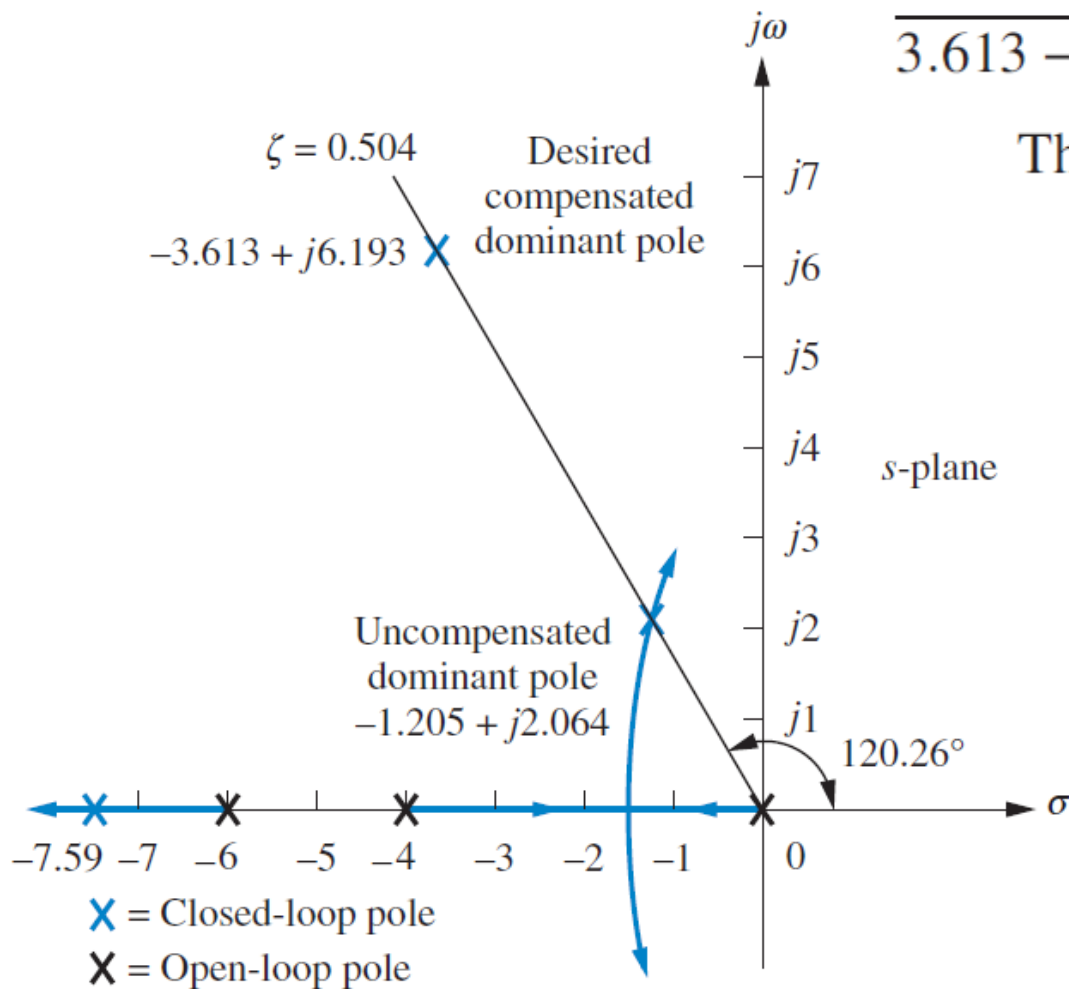
Thus, the imaginary part is  $\omega_d = 3.613 \tan(180^\circ - 120.26^\circ) = 6.193$

Next we design the location of the compensator zero.

# Improving Transient Response via Cascade Compensation

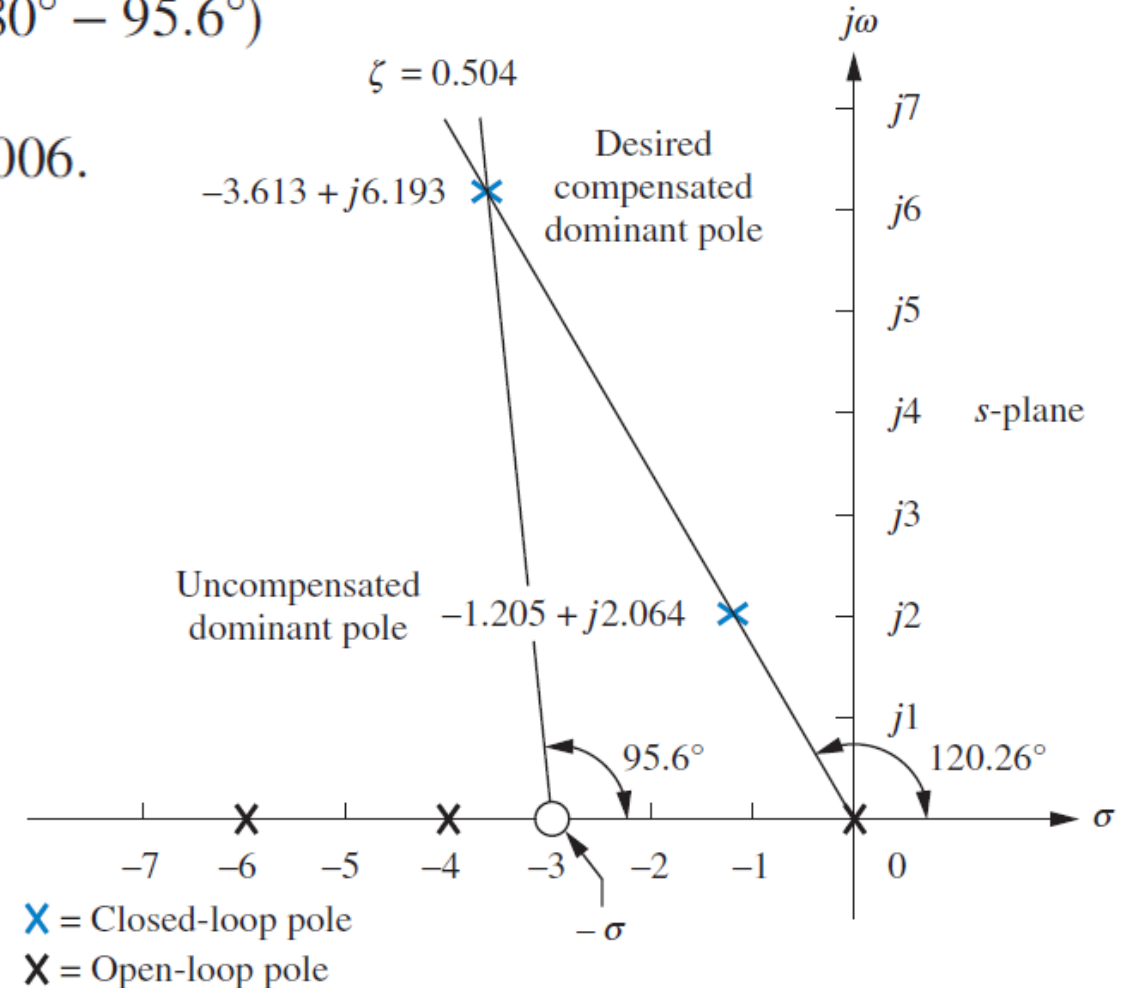
## Ideal Derivative Compensation (PD)

Soln. Cont.



$$\frac{6.193}{3.613 - \sigma} = \tan(180^\circ - 95.6^\circ)$$

Thus,  $\sigma = 3.006$ .



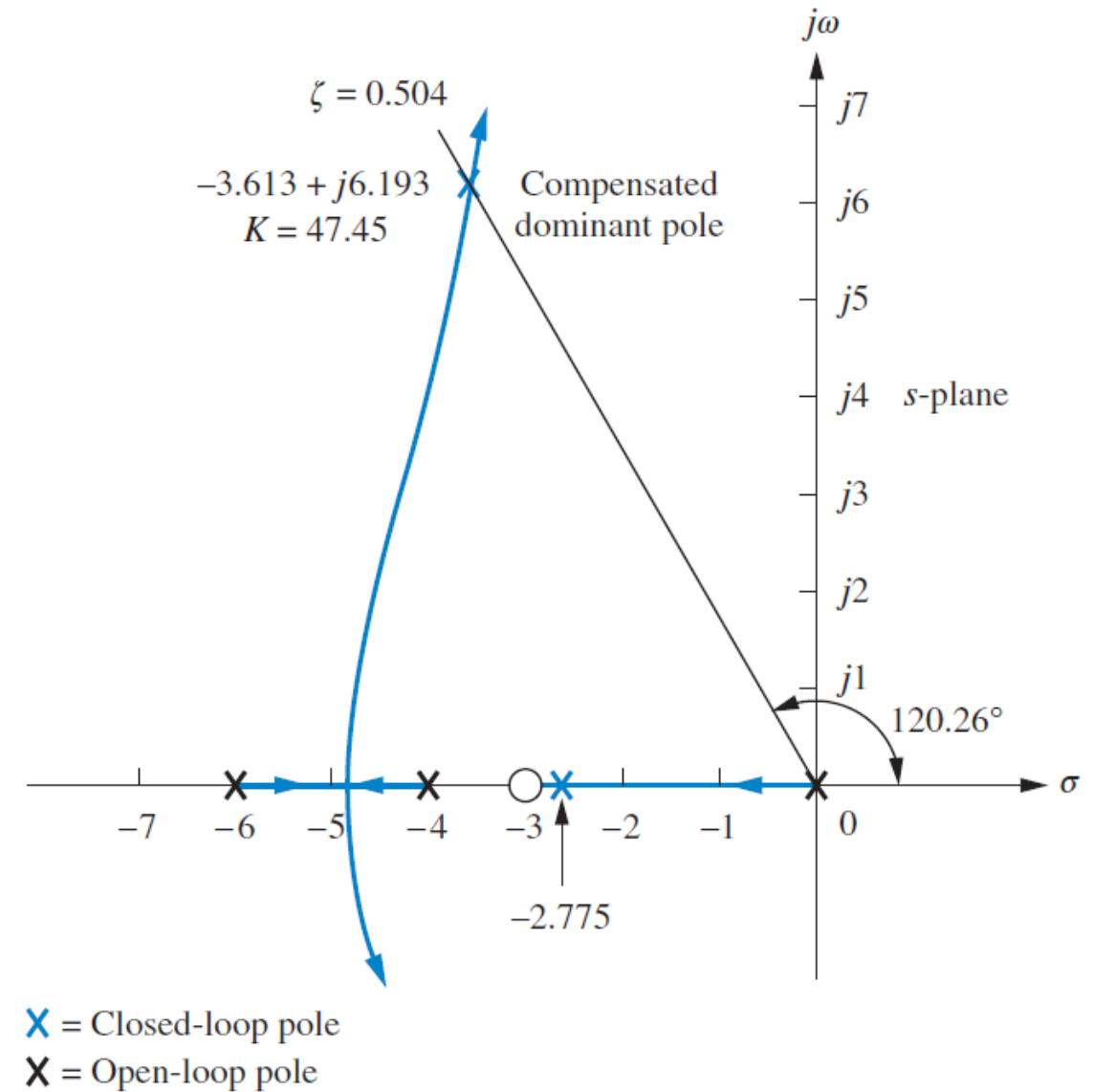
# Improving Transient Response via Cascade Compensation

## Ideal Derivative Compensation (PD)

### Soln. Cont.

For the uncompensated system, the estimate of the transient response is accurate since the third pole is at least five times the real part of the dominant, second-order pair.

The second-order approximation for the compensated system, however, may be invalid because there is no approximate closed-loop third-pole and zero cancellation.

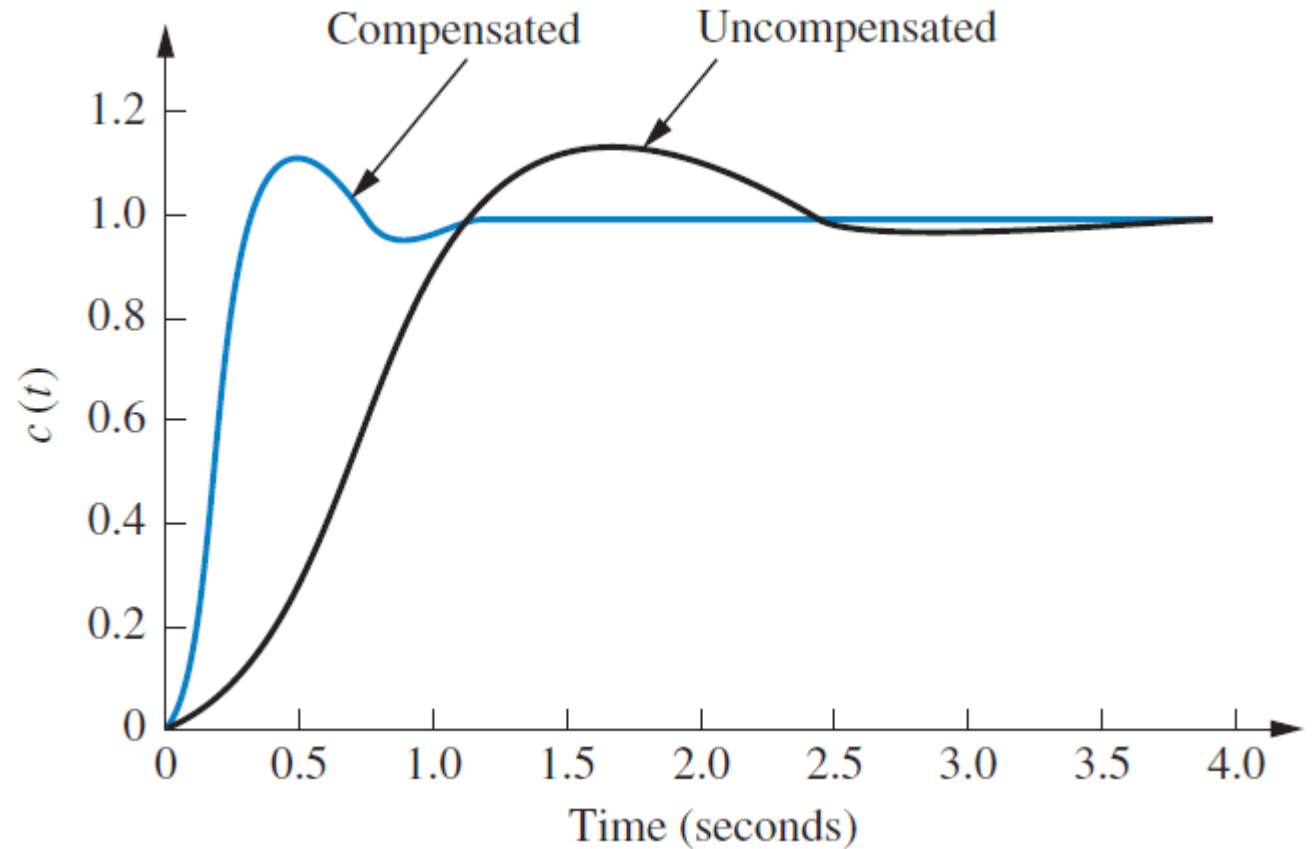


# Improving Transient Response via Cascade Compensation

## Ideal Derivative Compensation (PD)

### Soln. Cont.

A **simulation** or a **partial-fraction** expansion of the closed-loop response to compare the residue of the pole at  $-2.775$  to the residues of the dominant poles at  $-3.613 \pm j6.193$  is required.

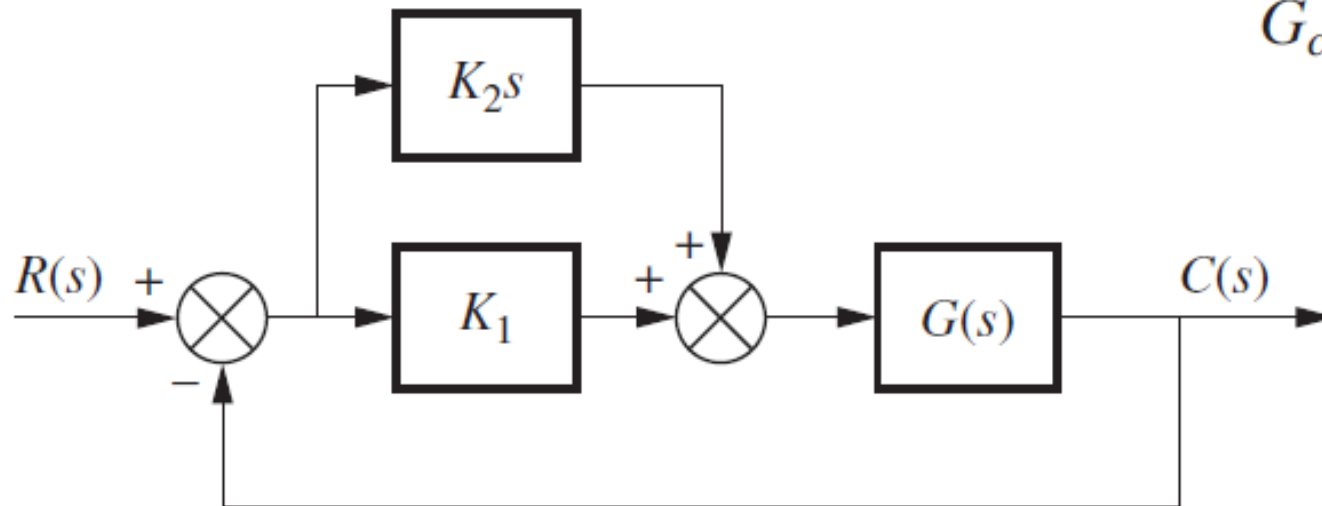




# Improving Transient Response via Cascade Compensation

## Ideal Derivative Compensation (PD) - Implementation Method

- The implementation consists of feeding the error (proportional) plus the derivative of the error forward to the plant.
- $K_1/K_2$  is chosen to equal the negative of the compensator zero, and  $K_2$  is chosen to contribute to the required loop-gain value.



$$G_c(s) = K_2s + K_1 = K_2 \left( s + \frac{K_1}{K_2} \right)$$

# Improving Steady-State Error and Transient Response

- Basically, we first improve the transient response, and then we improve the steady-state error of this compensated system. A disadvantage of this approach is the slight decrease in the speed of the response when the steady-state error is improved.
- As an alternative, we can improve the steady-state error first and then follow with the design to improve the transient response.
  - A disadvantage of this approach is that the improvement in transient response in some cases yields deterioration in the improvement of the steady-state error, which was designed first.
  - In other cases, the improvement in transient response yields further improvement in steady-state errors. Thus, a system can be overdesigned with respect to steady-state errors. Overdesign is usually not a problem unless it affects cost or produces other design problems.

Here, we first design for transient response and then design for steady-state error.

# Improving Steady-State Error and Transient Response

## PID Controller Design

Its transfer function is

$$G_c(s) = K_1 + \frac{K_2}{s} + K_3s = \frac{K_1s + K_2 + K_3s^2}{s} = \frac{K_3 \left( s^2 + \frac{K_1}{K_3}s + \frac{K_2}{K_3} \right)}{s}$$

which has two zeros plus a pole at the origin.

- One zero and the pole at the origin can be designed as the ideal integral compensator;
- The other zero can be designed as the ideal derivative compensator.

# Improving Steady-State Error and Transient Response

## PID Controller Design

The design technique consists of the following steps:

1. Evaluate the performance of the uncompensated system to determine how much improvement in transient response is required.
2. Design the PD controller to meet the transient response specifications. The design includes the zero location and the loop gain.
3. Simulate the system to be sure all requirements have been met.
4. Redesign if the simulation shows that requirements have not been met.
5. Design the PI controller to yield the required steady-state error.
6. Determine the gains,  $K_1$ ,  $K_2$ , and  $K_3$ .
7. Simulate the system to be sure all requirements have been met.
8. Redesign if simulation shows that requirements have not been met.

# Cascade Compensators

| Function                   | Compensator | Transfer function     | Characteristics                                                                                                                                                                                                                                                                  |
|----------------------------|-------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Improve steady-state error | PI          | $K \frac{s + z_c}{s}$ | <ol style="list-style-type: none"><li>1. Increases system type.</li><li>2. Error becomes zero.</li><li>3. Zero at <math>-z_c</math> is small and negative.</li><li>4. Active circuits are required to implement.</li></ol>                                                       |
| Improve transient response | PD          | $K(s + z_c)$          | <ol style="list-style-type: none"><li>1. Zero at <math>-z_c</math> is selected to put design point on root locus.</li><li>2. Active circuits are required to implement.</li><li>3. Can cause noise and saturation; implement with rate feedback or with a pole (lead).</li></ol> |

# Cascade Compensators

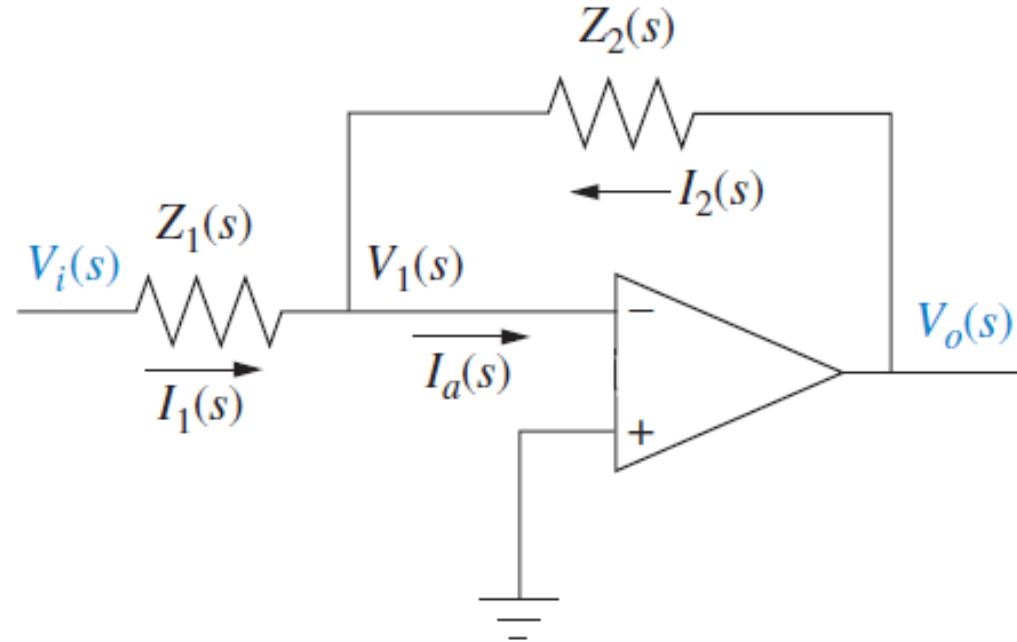
| Function                                          | Compensator | Transfer function                                       | Characteristics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------|-------------|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Improve steady-state error and transient response | PID         | $K \frac{(s + z_{\text{lag}})(s + z_{\text{lead}})}{s}$ | <ol style="list-style-type: none"><li>1. Lag zero at <math>-z_{\text{lag}}</math> and pole at origin improve steady-state error.</li><li>2. Lead zero at <math>-z_{\text{lead}}</math> improves transient response.</li><li>3. Lag zero at <math>-z_{\text{lag}}</math> is close to, and to the left of, the origin.</li><li>4. Lead zero at <math>-z_{\text{lead}}</math> is selected to put design point on root locus.</li><li>5. Active circuits required to implement.</li><li>6. Can cause noise and saturation; implement with rate feedback or with an additional pole.</li></ol> |

# Physical Realization of Compensation

## Active-Circuit Realization

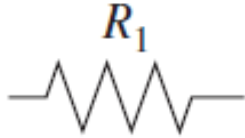
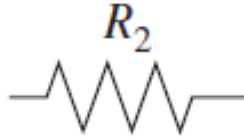
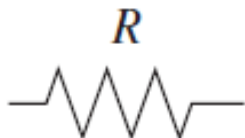
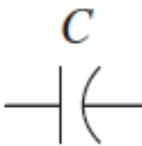
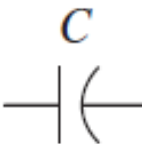
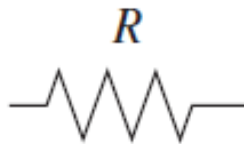
Recall.

$$\frac{V_o(s)}{V_i(s)} = -\frac{Z_2(s)}{Z_1(s)}$$



# Physical Realization of Compensation

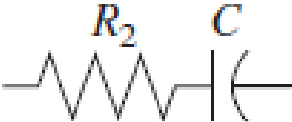
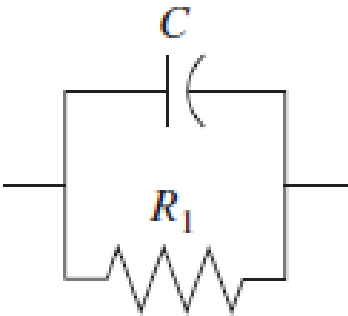
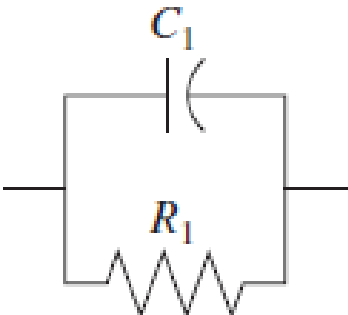
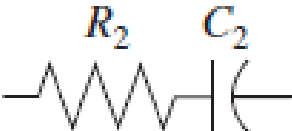
## Active-Circuit Realization

| Function        | $Z_1(s)$                                                                            | $Z_2(s)$                                                                              | $G_c(s) = -\frac{Z_2(s)}{Z_1(s)}$ |
|-----------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-----------------------------------|
| Gain            |    |    | $-\frac{R_2}{R_1}$                |
| Integration     |    |    | $-\frac{1}{RCs}$                  |
| Differentiation |  |  | $-RCs$                            |



# Physical Realization of Compensation

## Active-Circuit Realization

| Function       | $Z_1(s)$                                                                            | $Z_2(s)$                                                                              | $G_c(s) = -\frac{Z_2(s)}{Z_1(s)}$                                                                  |
|----------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| PI controller  |    |    | $-\frac{R_2}{R_1} \left( s + \frac{1}{R_2 C} \right)$                                              |
| PD controller  |   |    | $-R_2 C \left( s + \frac{1}{R_1 C} \right)$                                                        |
| PID controller |  |  | $-\left[ \left( \frac{R_2}{R_1} + \frac{C_1}{C_2} \right) + R_2 C_1 s + \frac{R_1 C_2}{s} \right]$ |

# Physical Realization of Compensation

**Ex.** Implement the given PID controller

**Soln.** It can be put in the form

$$G_c(s) = s + 56.42 + \frac{27.96}{s}$$

$$G_c(s) = \frac{(s + 55.92)(s + 0.5)}{s}$$

Comparing the PID controller in Table

$$\frac{R_2}{R_1} + \frac{C_1}{C_2} = 56.42 \quad R_2 C_1 = 1 \quad \frac{1}{R_1 C_2} = 27.96$$

Since there are four unknowns and three equations, we **arbitrarily** select a practical value for one of the elements.

Selecting  $C_2 = 0.1 \mu\text{F}$ , the remaining values are found to be  $R_1 = 357.65 \text{ k}\Omega$ ,  $R_2 = 178,891 \text{ k}\Omega$ , and  $C_1 = 5.59 \mu\text{F}$ .

The circuit element values have been **rounded off**.

